Abstract Algebra for Computer Sciences

MATHEMATICS DEPARTMENT

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Course Name: **Abstract Algebra for CS.**

Course Code: **(Math 333 [2L + 2S])**

Course Grade: **150 = 37 Course Work + 8 Oral Exam + Final Exam 105**

Course Contents

1- **Basic definitions of groups (Group, finite group, order of group, subgroup, cyclic groups)**
2- **Rings**
3- **Fields**

37 Grade (Course Work for Part 1)

Component	Grades
Quiz in Lecture 4	10
Assignments	7
Midterm Exam in Lecture 6	20

1. Abstract Algebra — "Group Theory"

What is abstract Algebra? What is a Group?

Before we give the definition of a group, let's see a few examples.

Example 1: "Clock Arithmetic" or "Modular Arithmetic"

Let Clock Numbers: 0, 1, 2, 3, 4, 5, 6

+	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	2	3	4	5	6	0
2	2	3	4	5	6	0	1
3	3	4	5	6	0	1	2
4	4	5	6	0	1	2	3
5	5	6	0	1	2	3	4
6	6	0	1	2	3	4	5

Example 2: The set of all integers $\mathbb{Z}$ with the usual addition

Thus, we have the following comparison:

	Clock Arithmetic	The set of all integers $\mathbb{Z}$
Elements	S = {0,1,2,3,4,5,6}	$\mathbb{Z}$ = {…,-2,-1,0,1,2,…}
Operation	+ mod 7	+
Set closed under operation	Yes	Yes
Associative	Yes	Yes
Identity	0	0
Inverses	Exist for every element	Exist for every element

Thus, the mathematicians abstract these properties to give the definition of the group.

Binary Operations

Definition: Let S be a nonempty set. A binary operation $(*)$ on S is a function from $S \times S$ into S.

$$* : S \times S \to S$$
$$(x, y) \mapsto x * y$$

Since the image of $(*)$ is a subset of S, we say that S **is closed under $(*)$.**

Examples:

1. The set of integers $\mathbb{Z}$ is closed under $(+)$, $(-)$, and $(\times)$.

   As, $\forall\, x, y \in \mathbb{Z}$, $x + y, x - y, x \cdot y \in \mathbb{Z}$.

2. The set of natural numbers $\mathbb{N}$ is closed under $(+)$ but it is not closed under $(-)$.

   As, $3, 5 \in \mathbb{N}$ but $3 - 5 = -2 \notin \mathbb{N}$.

Definition of a Group

Definition: A group is an ordered pair $(G, *)$, where G is a nonempty set and $*$ is a binary operation on G such that the following properties hold:

(G1) $\forall\, a, b, c \in G,\ a * (b * c) = (a * b) * c$ **(associative law).**
(G2) $\exists\, e \in G\ \ \forall\, a \in G:\ a * e = a = e * a$ **(existence of an identity).**
(G3) $\forall\, a \in G\ \ \exists\, b \in G:\ a * b = e = b * a$ **(existence of an inverse).**

Definition: If $(G, *)$ is a group and $a * b = b * a$ for all $a, b \in G$, then the group is called an abelian group (commutative group).

Note that:

1. A non-empty set S with a binary operation '*' is called a semigroup if it satisfies the following conditions:

(i) '*' is closed **(ii) '*' is associative**

3. A monoid is a semigroup with an identity element.

Examples of Groups

1. ($\mathbb{Z}$, +), where + is the usual addition of integer numbers.

- **$\forall$ a, b, c $\in \mathbb{Z}$, (a + b) + c = a + (b + c).**
- **$\exists$ 0 $\in \mathbb{Z}$: $\forall$ a $\in \mathbb{Z}$, a + 0 = 0 + a = a.**
- **$\forall$ a $\in \mathbb{Z}$ $\exists$ −a: a + (−a) = (−a) + a = 0.**

Thus, ($\mathbb{Z}$, +) is a group. And $\forall$ a, b $\in \mathbb{Z}$, a + b = b + a. Thus ($\mathbb{Z}$, +) is an abelian group.

❖ We can show that ($\mathbb{Q}$, +), ($\mathbb{R}$, +), ($\mathbb{C}$, +) are also abelian groups, where + is the usual addition.

2. ($\mathbb{Q}\backslash\{0\}$, ·), where '·' is the usual multiplication.

- **$\forall$ a, b, c $\in \mathbb{Q}\backslash\{0\}$, (a·b)·c = a·(b·c).**
- **$\exists$ 1 $\in \mathbb{Q}\backslash\{0\}$: $\forall$ a $\in \mathbb{Q}\backslash\{0\}$, a·1 = 1·a = a.**
- **$\forall$ a $\in \mathbb{Q}\backslash\{0\}$ $\exists$ 1/a: a·(1/a) = (1/a)·a = 1.**

Thus, ($\mathbb{Q}\backslash\{0\}$, ·) is a group.

And $\forall$ a, b $\in \mathbb{Q}\backslash\{0\}$, a·b = b·a. Thus ($\mathbb{Q}\backslash\{0\}$, ·) is an abelian group.

❖ We can show that ($\mathbb{R}\backslash\{0\}$, ·), ($\mathbb{C}\backslash\{0\}$, ·) are also abelian groups, where '·' is the usual multiplication.

3. Let M(2, $\mathbb{R}$) be the set of all 2×2 matrices over $\mathbb{R}$.

$$M(2,\mathbb{R}) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : \ a,b,c,d \ \in \ \mathbb{R} \right\}$$

With '+' the usual addition of matrices and '×' the usual multiplication of matrices,

(M(2,$\mathbb{R}$), +) is a commutative group,

but (M(2,$\mathbb{R}$), ×) is not a group, as not every square matrix has an inverse.

For example, let $A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$, then A has no inverse as |A| = 0.

4. Let GL(2, $\mathbb{R}$) be the set of all 2×2 matrices over $\mathbb{R}$ with det(A) nonzero.

$$GL(2,\mathbb{R}) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : \ a,b,c,d \ \in \ \mathbb{R}, |A| \neq \ 0 \right\}$$

Then, (GL(2,ℝ), ×) is a group but it is not commutative. Indeed, let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, B = \begin{bmatrix} l & m \\ n & p \end{bmatrix}, C = \begin{bmatrix} r & s \\ t & v \end{bmatrix}$$

1. $A \times B = \begin{bmatrix} al + bn & am + bp \\ cl + dn & cm + dp \end{bmatrix} \in GL(2, \mathbb{R})$

2. (A × B) × C = A × (B × C) (verify)

3. There exists an identity $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$: ∀ A ∈ GL(2,ℝ), A × I_2 = I_2 × A = A

4. $\forall A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in$ GL(2,ℝ) ∃ $A^{-1} = A^{-1} = \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$: $A A^{-1} = A^{-1}A = I_2$

And multiplication of matrices is not commutative.

For example, let $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix}$, then

$$AB = \begin{bmatrix} 8 & 1 \\ 9 & 0 \end{bmatrix} \text{ and } BA = \begin{bmatrix} 2 & 7 \\ 3 & 6 \end{bmatrix}, \text{ thus AB} \neq \text{BA.}$$

This group is called the general linear group.

❖ **Similarly, we can prove that (SL(2,ℝ), ×) is a group but it is not commutative, where**

$$SL(2, \mathbb{R}) = \{ [\, a \;\; b \,;\, c \;\; d \,] : a, b, c, d \in \mathbb{R}, |A| = 1 \} \quad \textit{(verify)}$$

This group is called the special linear group.

Theorem (Uniqueness of Identity and Inverse)

Theorem: Let (G, ∗) be a group.
(i) The identity element is unique.
(ii) For all a ∈ G, there exists a unique inverse.

Proof:

(i) Let e, e_1 ∈ G be two identities, then

$$e_1 = e_1 * e = e.$$

Therefore, the identity element is unique.

(ii) Let a ∈ G. Since (G, ∗) is a group, there exists b ∈ G such that a ∗ b = e = b ∗ a.

Suppose there exists another inverse c ∈ G for a. We will show that b = c. Now

$$b = b * e = b * (a * c) \quad \text{[substituting } e = a * c\text{]}$$

$$= (b * a) * c \quad \text{[associativity of } *\text{]}$$

$$= e * c \quad \text{[because } b * a = e\text{]}$$

$$= c.$$

Thus, the inverse element b is unique (denoted by a^{-1}).

Theorem (Inverse Properties and Cancellation Law)

Theorem: Let $(G, *)$ be a group.

1. $(a^{-1})^{-1} = a$ for all $a \in G$.
2. $(a * b)^{-1} = b^{-1} * a^{-1}$ for all $a, b \in G$.
3. (Cancellation Law) For all $a, b, c \in G$, if either $a * c = b * c$ or $c * a = c * b$, then $a = b$.

Proof:

1. Let $a \in G$. Then

$$a^{-1} * (a^{-1})^{-1} = e$$

$$\Rightarrow a * a^{-1} * (a^{-1})^{-1} = a \quad \text{[multiplying by a on the left]}$$

$$\Rightarrow (a * a^{-1}) * (a^{-1})^{-1} = a$$

$$\Rightarrow e * (a^{-1})^{-1} = a$$

$$\Rightarrow (a^{-1})^{-1} = a$$

OR,

$$a^{-1} * a = a * a^{-1} = e, thus\ (a^{-1})^{-1} = a$$

2. Let $a, b \in G$. Then

$$(a * b) * (a * b)^{-1} = e$$

$$\Rightarrow a^{-1} * (a * b) * (a * b)^{-1} = a^{-1} * e$$

$$\Rightarrow (a^{-1} * a) * b * (a * b)^{-1} = a^{-1}$$

$$\Rightarrow e * b * (a * b)^{-1} = a^{-1}$$

$$\Rightarrow b * (a * b)^{-1} = a^{-1}$$

$$\Rightarrow b^{-1} * b * (a * b)^{-1} = b^{-1} * a^{-1}$$

$$\Rightarrow e * (a * b)^{-1} = b^{-1} * a^{-1}$$

$$\Rightarrow (a * b)^{-1} = b^{-1} * a^{-1}$$

OR,

$$a * b * (b^{-1} * a^{-1}) = a * (b * b^{-1}) * a^{-1} = (a * e) * a^{-1} = a * a^{-1} = e$$

And,

$$(b^{-1} * a^{-1}) * a * b = b^{-1} * (a^{-1} * a) * b = (b^{-1} * e) * b = b^{-1} * b = e$$

Thus, $(a * b)^{-1} = b^{-1} * a^{-1}$.

3. Let a, b, c ∈ G. Suppose $a * c = b * c$. Now

$$(a * c) * c^{-1} = (b * c) * c^{-1}$$

$$\Rightarrow a * (c * c^{-1}) = b * (c * c^{-1}) \quad \text{[associativity]}$$

$$\Rightarrow a * e = b * e \quad \text{[since } c * c^{-1} = e\text{]}$$

$$\Rightarrow a = b.$$

Similarly, if $c * a = c * b$, then we can show that a = b.

Question: Let $G = \{(a, b) \mid a, b \in \mathbb{R}, a \neq 0\} = \mathbb{R}\backslash\{0\} \times \mathbb{R}$. Define a binary operation $*$ on G by

$$(a, b) * (c, d) = (ac, b + d) \quad \text{for all } (a, b), (c, d) \in G.$$

Show that $(G, *)$ is an abelian group.

Solution:

- **'*' is well-defined:**

Let $((a,b),(l,m)) = ((c,d),(n,p)), then\ (a,b) = (c,d) and\ (l,m) = (n,p),$

$$thus\ a = c, b = d,\ l = n, m = p.$$

$$We\ want\ to\ show\ that\ (a,b) * (l,m) = (c,d) * (n,p).$$

$$(a,b) * (l,m) = (al, b+m) = (cn, d+p) = (c,d) * (n,p).$$

- **'*' is Binary:**

Let $(a,b),(c,d) \in G. Then\ a \neq 0\ and\ c \neq 0, so\ ac \neq 0.$

Thus, $(a,b) * (c,d) = (ac, b+d) \in G. Hence, G\ is\ closed\ under\ *.$

- **'*' is Associative:**

Let $(a,b),(c,d),(l,m) \in G, then$

$$((a,b) * (c,d)) * (l,m) = (ac, b+d) * (l,m) = (acl, b+d+m)$$

$$= (a,b) * (cl, d+m) = (a,b) * ((c,d) * (l,m)).$$

Thus, $*$ is associative.

- **Existence of identity:**

We search for $(c,d) \in G \; such \; that$

$$(a,b) * (c,d) = (c,d) * (a,b) = (a,b) \; for \; all \; (a,b) \in G.$$

$$(a,b) * (c,d) = (ac, b+d) = (a,b)$$

Thus, $ac = a \; and \; b+d = b, so \; c = 1 \; and \; d = 0.$

Therefore, $(c,d) = (1,0) \in G \; is \; the \; identity \; element.$

- **Existence of inverse:**

Let $(a,b) \in G. We \; search \; for \; (c,d) \in G \; such \; that$

$$(a,b) * (c,d) = (c,d) * (a,b) = (1,0).$$

$$(a,b) * (c,d) = (ac, b+d) = (1,0)$$

Thus, $ac = 1 \; and \; b+d = 0, so \; c = \frac{1}{a} and \; d = -b.$

Therefore, $(c,d) = (1/a, -b) \in G \; is \; the \; inverse \; of \; (a,b).$

- **G is abelian:**

Let $(a,b),(c,d) \in G, then$

$$(a,b) * (c,d) = (ac, b+d) = (ca, d+b) = (c,d) * (a,b).$$

Thus, $*$ is commutative, and (G, $*$) is an abelian group. ■

2. Relations and Equivalence Classes

Definition: A relation R from a set A into a set B is a subset of A × B.

Definition: Let R be a binary relation on a set A. Then R is called

- **reflexive** if for all x ∈ A, x R x,
- **symmetric** if for all x, y ∈ A, x R y implies y R x,
- transitive if for all x, y, z ∈ A, x R y and y R z imply x R z.

Definition: A relation R on a set A is called **an equivalence relation** on A if R is reflexive, symmetric, and transitive.

Definition: Let R be an equivalence relation on a set A. For all $x \in A$, let [x] denote the set $[x] = \{y \in A \mid y \, R \, x\}$. The set [x] is called the equivalence class (with respect to R) determined by x.

Note that:

1. $A = \bigcup_{x \in a} [x]$
2. $\forall$ x, y ∈ A, either [x] = [y] or $[x] \cap [y] = \emptyset$.
3. If [x] = [y], then x R y.

Congruence Modulo n

Definition: The relation ($\equiv \bmod n$), $n \in \mathbb{Z}^+$ defined on $\mathbb{Z}$ as follows:

$$\forall\, a, b \in \mathbb{Z},\ a \equiv b \bmod n \iff n \mid (a - b) \iff (a - b)/n \in \mathbb{Z}.$$

Examples:

1. $3 \equiv 8 \bmod 5$, as $(3-8)/5 = -1 \in \mathbb{Z}$
2. $11 \equiv 2 \bmod 3$, as $(11-2)/3 = 3 \in \mathbb{Z}$
3. $15 \not\equiv 1 \bmod 4$, as $(15-1)/4 \notin \mathbb{Z}$

The relation $R = \{(a,b) \in \mathbb{Z}\times\mathbb{Z} : a \equiv b \bmod n, n \in \mathbb{Z}^+\}$ is an equivalence relation on $\mathbb{Z}$.

Proof:

1. $\forall\, a \in \mathbb{Z}, (a - a)/n = 0 \in \mathbb{Z} \Rightarrow a \equiv a \bmod n.$

   Then (a,a) ∈ R, thus R is reflexive.

2. Let $a, b \in \mathbb{Z}$: $(a, b) \in R \Rightarrow a \equiv b \bmod n$

$$\Rightarrow \frac{a-b}{n} = t \in \mathbb{Z}$$

$$\Rightarrow \frac{b-a}{n} = -t \in \mathbb{Z}$$

$$\Rightarrow b \equiv a \bmod n.$$

Then (b,a) ∈ R, thus R is symmetric.

3. Let $a, b, c \in \mathbb{Z}$: $(a, b) \in R \wedge (b, c) \in R$

$$\Rightarrow a \equiv b \bmod n \wedge b \equiv c \bmod n$$

$$\Rightarrow \frac{a-b}{n} = r \in \mathbb{Z} \wedge \frac{b-c}{n} = s \in \mathbb{Z}$$

$$\Rightarrow (a-c)/n = r + s \in \mathbb{Z}.$$

Then a $\equiv$ $c \bmod n$, thus $(a, c) \in R$. Therefore, R is transitive.

From 1, 2, and 3, we have R is an equivalence relation.

The equivalence classes of this relation form a partition on $\mathbb{Z}$. Let

$$\mathbb{Z}_n = \{[a] : a \in \mathbb{Z}\} \text{ and } [a] = \{x \in \mathbb{Z} : x \equiv a \bmod n\}$$

In fact, we have exactly n distinct equivalence classes, namely

$$\mathbb{Z}_n = \{[0], [1], \dots, [n-1]\}.$$

Example: n = 5

($\mathbb{Z}_5$, +) is a Commutative Group

Prove that: ($\mathbb{Z}_5$, +) is a commutative group, where [a] + [b] = [a+b].

Proof:

1. We prove that the operation is **well defined:**

Let $([a], [b]) = ([a_1], [b_1])$,

$$then\ [a] = [a_1]\ and\ \ [b] = [b_1].$$

We go to prove that $[a] + [b] = [a_1] + [b_1]$.

$$[a] = [a_1] \Rightarrow a \equiv a_1 \bmod 5 \Rightarrow (a - a_1)/5 = t \in \mathbb{Z} \Rightarrow a = 5t + a_1, t \in \mathbb{Z}.$$

$$[b] = [b_1] \Rightarrow b \equiv b_1 \bmod 5 \Rightarrow (b - b_1)/5 = s \in \mathbb{Z} \Rightarrow b = 5s + b_1, s \in \mathbb{Z}.$$

$$[a] + [b] = [a + b] = [5t + a_1 + 5s + b_1] = [a_1 + b_1 + 5(t + s)]$$

$$= [a_1 + b_1] + [5(t + s)] = [a_1 + b_1] = [a_1] + [b_1].$$

Cayley Table for ($\mathbb{Z}_5$, +):

+	[0]	[1]	[2]	[3]	[4]
[0]	[0]	[1]	[2]	[3]	[4]
[1]	[1]	[2]	[3]	[4]	[0]
[2]	[2]	[3]	[4]	[0]	[1]
[3]	[3]	[4]	[0]	[1]	[2]
[4]	[4]	[0]	[1]	[2]	[3]

From the Cayley table we have:

1. $\forall\, [a], [b] \in \mathbb{Z}_5, [a] + [b] \in \mathbb{Z}_5$. Thus + is a binary operation.

2. $\forall\, [a], [b], [c] \in \mathbb{Z}_5,$

$$[a] + ([b] + [c]) = [a] + [b + c] = [a + (b + c)] = [(a + b) + c]$$
$$= [(a + b)] + [c] = ([a] + [b]) + [c].$$

Thus + is associative.

3. [0] is the identity element.

4. Table of inverses:

Element	[0]	[1]	[2]	[3]	[4]
Inverse	[0]	[4]	[3]	[2]	[1]

4. The table is symmetric around the main diagonal, so $\mathbb{Z}_5$ is abelian, i.e.,

$\forall\, [a], [b] \in \mathbb{Z}_5, [a] + [b] = [a + b] = [b + a] = [b] + [a]$. Thus + is commutative.

Therefore, ($\mathbb{Z}_5$, +) is a commutative group. ■